UNIVERSITY OF JEDDAH

COLLEGE OF COMPUTER SCIENCE AND ENGINEERING

SOFTWARE ENGINEERING DEPT.

NUTOQ

A web-based platform tailored for adults individuals who stutter.

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Abstract

Stuttering is a speech disorder that affects fluency and communication, often leading to social and emotional challenges for adults. This project presents the design and implementation of a web-based platform tailored for adults who stutter, aimed at providing an accessible and interactive environment for speech improvement. The platform features a user-friendly interface to ensure ease of navigation and engagement. It incorporates an adaptive training mechanism that displays words and sentences according to the user's proficiency level, supported by a progressive level system that gradually advances from simple words to complex sentences. To enhance learning, speech-to-text (STT) technology is integrated for real-time evaluation of pronunciation accuracy, automatically progressing users upon successful completion of exercises. Additionally, the system offers immediate supportive hints, such as reminders to breathe or speak slowly, whenever errors or pauses occur, fostering a positive and guided experience. A progress dashboard is included to visualize user performance and current level, promoting motivation and continuous improvement. This platform aims to leverage technology to provide personalized, convenient, and effective support for adults managing stuttering, contributing to improved speech fluency and confidence.

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CHAPTER 1
PLANNING

1.1 Description of the context (Scope) and the problem to solve

Stuttering is one of the most common speech disorders, directly affecting fluency and self-confidence. Many individuals struggle to maintain consistent practice outside therapy sessions due to the lack of simple interactive tools for daily self-training.

In addition, some patients do not have access to speech therapy centers in their area or cannot attend sessions regularly due to work or study commitments. Others may feel socially embarrassed to visit clinics and participate in therapy, which increases isolation and lowers improvement chances. Furthermore, the high cost of therapy sessions in some centers makes them unaffordable for many patients.

The core problem: people who stutter need a way to practice speaking interactively, with an automatic evaluation mechanism using Speech-to-Text (STT). Patients and consultants also need a simple way to monitor progress through the same account, without separate roles or complex systems.

1.2 Aim and Objectives

1.2.1 Aim:

The aim of this project is to develop an interactive web-based platform to support adults' individuals who stutter by enabling them to practice daily speech exercises in a simple, self-guided manner. The proposed platform will provide automatic performance evaluation using Speech-to-Text (STT) technology and present progress clearly, accessible by both the patient and the consultant.

The proposed platform will compare what the patient says with the displayed word/sentence. If correct, the user automatically advances to the next exercise; if incorrect, supportive hints appear (e.g., "take a breath," "try to slow down").

The proposed platform provides a practical and cost-effective solution for those unable to attend sessions, socially uncomfortable with therapy visits, or unable to afford long-term treatment fees.

1.2.2 Objectives:

1. To design and implement a user-friendly web-based interface that allows users to use the platform easily.
2. To provide an interactive training mechanism that displays words or sentences based on the user's level.
3. To integrate STT technology to evaluate pronunciation correctness and automatically advance to the next exercise upon success.
4. To offer immediate supportive hints (e.g., "take a breath," "speak slowly") when mistakes or pauses occur.
5. To develop a progressive level system starting with simple words and gradually advancing to sentences of higher difficulty.
6. To create a progress dashboard showing the current level.

1.3 Report outline

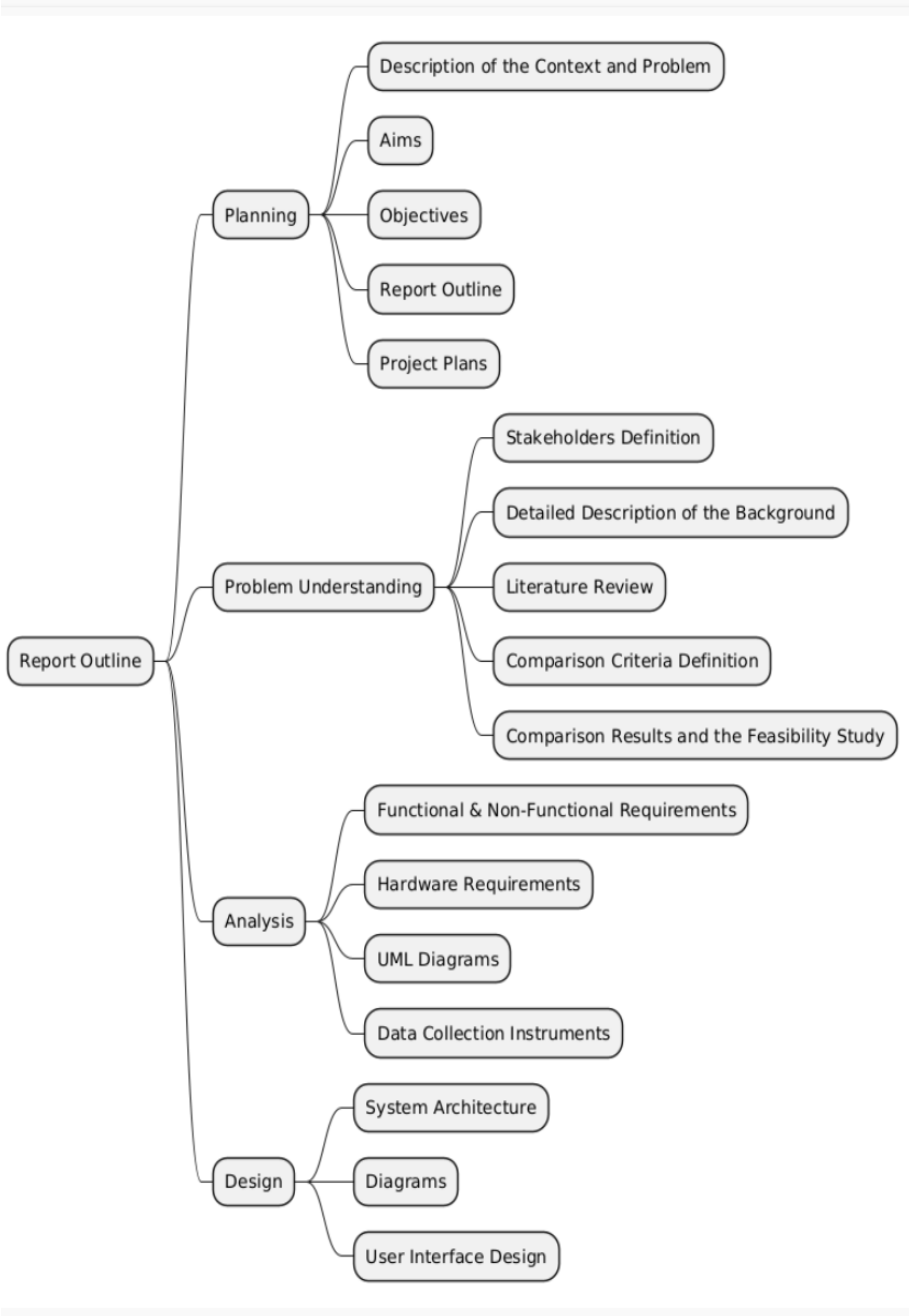
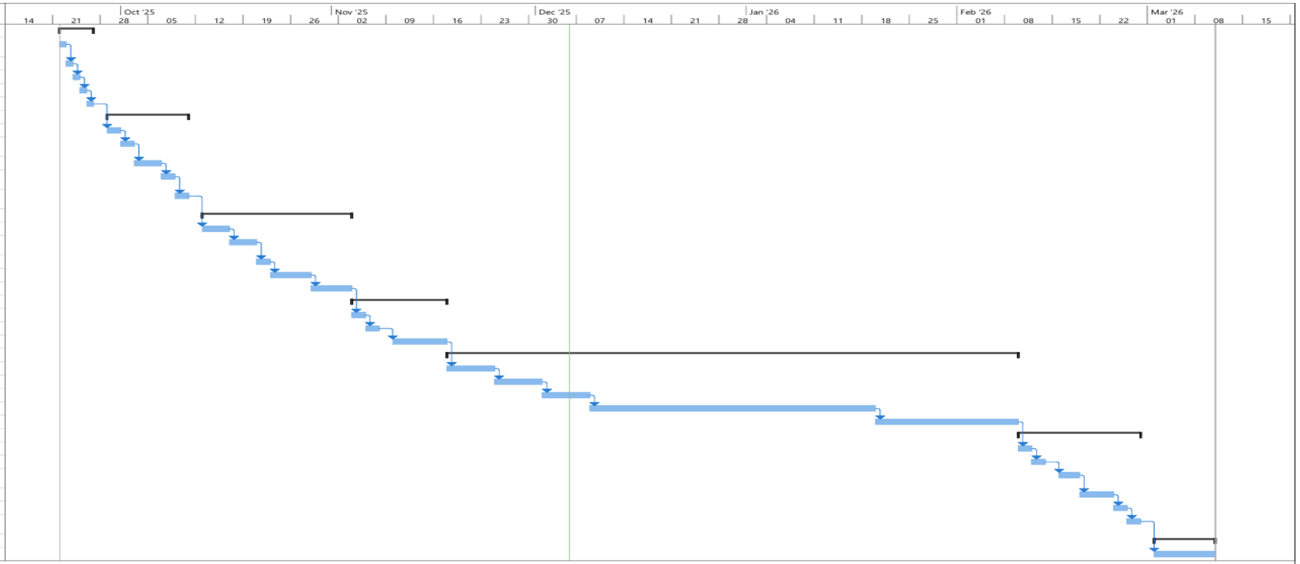


Figure 1.1 Report outline

1.4 Project plan

ID	ID	Task Mode	Task Name	Duration	Start	Finish	Predecessors
1	1		Planning	5 days	Mon 22-09-25	Fri 26-09-25	
2	2		Description of the context (Scope) and the problem	1 day	Mon 22-09-25	Mon 22-09-25	
3	3		Aims	1 day	Tue 23-09-25	Tue 23-09-25	2
4	4		Objectives	1 day	Wed 24-09-25	Wed 24-09-25	3
5	5		Report outline	1 day	Thu 25-09-25	Thu 25-09-25	4
6	6		Project Plan	1 day	Fri 26-09-25	Fri 26-09-25	5
7	7		Problem Understanding	10 days	Mon 29-09-25	Fri 10-10-25	
8	8		Stakeholders definition	2 days	Mon 29-09-25	Tue 30-09-25	6
9	9		Detailed Description of the Background	2 days	Wed 01-10-25	Thu 02-10-25	8
10	10		Literature Review	2 days	Fri 03-10-25	Mon 06-10-25	9
11	11		Comparison criteria definition	2 days	Tue 07-10-25	Wed 08-10-25	10
12	12		Comparison results and the feasibility study	2 days	Thu 09-10-25	Fri 10-10-25	11
13	13		Analysis	16 days	Mon 13-10-25	Mon 03-11-25	
14	14		Functional Requirements	4 days	Mon 13-10-25	Thu 16-10-25	12
15	15		Non-Functional Requirements	2 days	Fri 17-10-25	Mon 20-10-25	14
16	16		Hardware Requirements	2 days	Tue 21-10-25	Wed 22-10-25	15
17	17		UML diagrams	4 days	Thu 23-10-25	Tue 28-10-25	16
18	18		Data collection instrument	4 days	Wed 29-10-25	Mon 03-11-25	17
19	19		Design	10 days	Tue 04-11-25	Mon 17-11-25	
20	20		System Architecture	2 days	Tue 04-11-25	Wed 05-11-25	18
21	21		Diagrams	2 days	Thu 06-11-25	Fri 07-11-25	20
22	22		User Interface Design	6 days	Mon 10-11-25	Mon 17-11-25	21
23	23		Development	60 days	Tue 18-11-25	Mon 09-02-26	
24	24		Use the Design Document	5 days	Tue 18-11-25	Mon 24-11-25	22
25	25		APIs and Plugins	5 days	Tue 25-11-25	Mon 01-12-25	24
26	26		Data collection instrument	5 days	Tue 02-12-25	Mon 08-12-25	25
27	27		Implementation/Coding	30 days	Tue 09-12-25	Mon 19-01-26	26
28	28		Testing	15 days	Tue 20-01-26	Mon 09-02-26	27
29	29		Testing the developed artifact	14 days	Tue 10-02-26	Fri 27-02-26	
30	30		Conduct a testing study	2 days	Tue 10-02-26	Wed 11-02-26	28
31	31		Data collection	2 days	Thu 12-02-26	Fri 13-02-26	30
32	32		Clean the data and prepare for analysis	3 days	Mon 16-02-26	Wed 18-02-26	31
33	33		Analyze the collected data	3 days	Thu 19-02-26	Mon 23-02-26	32
34	34		Visualize the results	2 days	Tue 24-02-26	Wed 25-02-26	33
35	35		Understand what the results mean	2 days	Thu 26-02-26	Fri 27-02-26	34
36	36		Writing Final Report	7 days	Mon 02-03-26	Tue 10-03-26	
37	37		Writing Conclusion	7 days	Mon 02-03-26	Tue 10-03-26	35



Project: MS_Project
Date: Sat 06-12-25

Task | Split | Milestone | Summary | Project Summary | Inactive Task | Inactive Milestone | Inactive Summary | Manual Task | Duration-only | Manual Summary Rollup | Manual Summary | Start-only | Finish-only | External Tasks | External Milestone | Deadline | Progress | Manual Progress

Page 1

Figure 1.2 Project plan 1/2

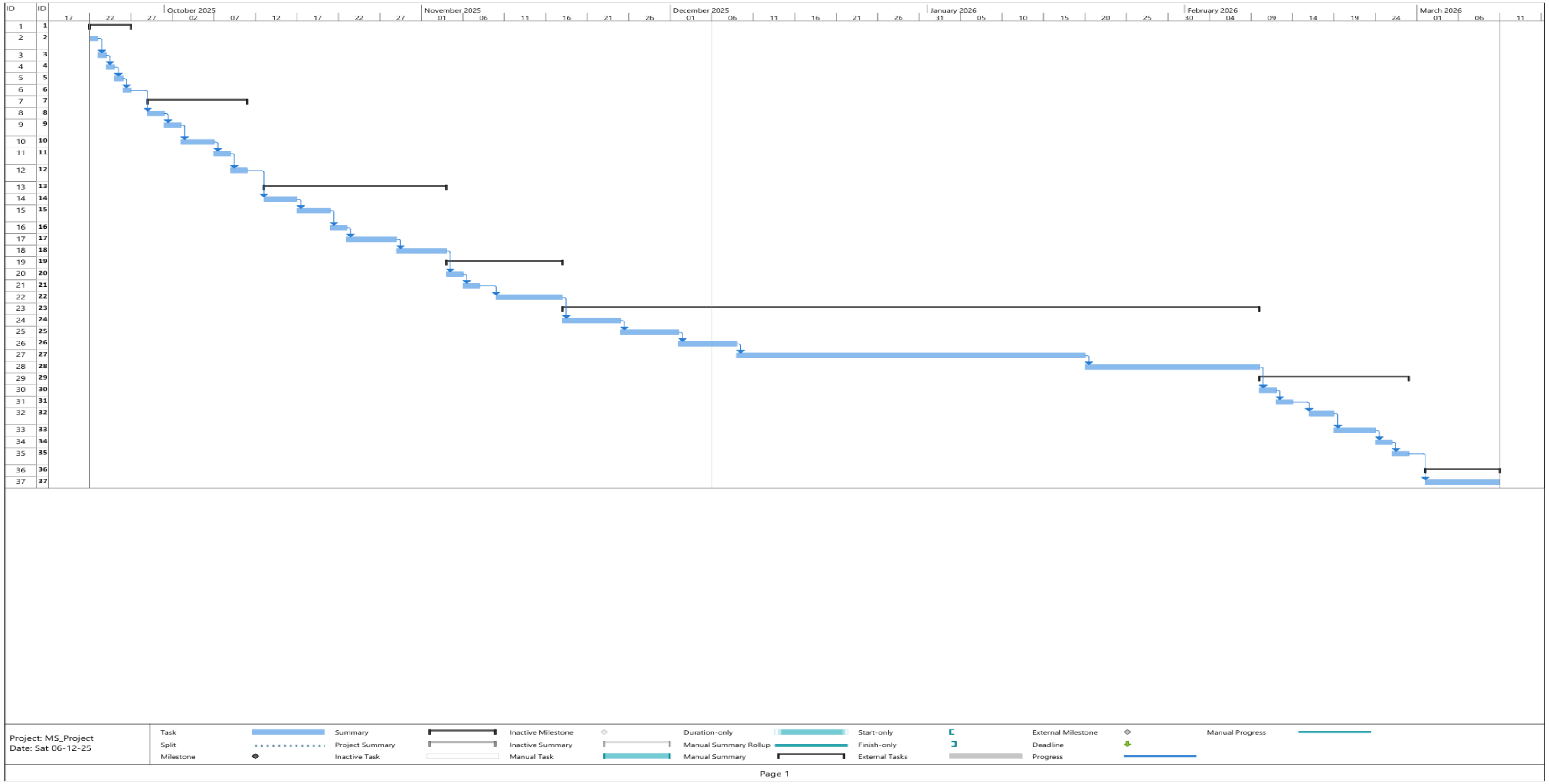


Figure 1.3 Project plan 2/2

CHAPTER 2
PROBLEM UNDERSTANDING

2.1 Stakeholders Definition:

The proposed platform primarily involves two main stakeholders: the patient (user) and the speech consultant. Each interacts with the platform in a complementary way to support speech improvement and consistent practice.

Patient (User): Individuals who stutter and use the system to practice speaking through interactive word and sentence exercises. The system evaluates their pronunciation using Speech-to-Text (STT) technology and automatically advances to the next exercise when a correct response is detected. Patients can also view their progress and improvement trends over time.

Speech Consultant: A speech specialist who can access the platform using the same account credentials as the patient, primarily to monitor the user's progress and observe performance data. The consultant's role focuses on supervision and follow-up. This step ensures that both stakeholders remain connected through one unified environment, allowing for convenient monitoring, transparency, and better understanding of user progress without the need for complex account management.

Contact person	Influence	Importance	Justification of Interests in the Project
Patient (User)	Medium	High	Main beneficiary of the platform. Uses the system daily to practice speech, receive STT evaluation, and track progress. Project success is measured mainly by how much it helps this stakeholder.
Speech Consultant	Medium	Low	Has professional expertise and can strongly influence how exercises, evaluation, and feedback are designed. Uses the platform to monitor patient progress and adjust therapy plans, so their input shapes system features and correctness.

Table 2.1 Stakeholders Definition

2.2 Detailed Description of the Background (Project Domain)

This project belongs to the domain of speech therapy and assistive technology, specifically focusing on the digital rehabilitation of individuals who stutter. Stuttering is a speech disorder that disrupts fluency through repetitions, prolongations, or involuntary pauses. Traditional therapy often requires consistent attendance at specialized clinics and personal guidance from speech therapists, which can be costly, time-consuming, and sometimes socially uncomfortable for patients.

With recent advancements in Artificial Intelligence (AI) and Speech-to-Text (STT) technologies, web-based platforms have become capable of providing automatic pronunciation feedback. This enables patients to practice independently, track progress, and maintain motivation between therapy sessions.

The proposed web-based platform leverages these technologies to create an accessible environment that allows users to train regularly without depending on clinical visits. Furthermore, the project lies within the growing field of digital health (e-health), which integrates modern web technologies with therapeutic approaches to improve accessibility and convenience.

The system aims to support individuals in Arabic-speaking regions, where accessible and culturally localized tools for speech improvement remain limited. By offering simple, browser-based training experience, the project combines accessibility, affordability, and continuous progress tracking in one integrated solution.

2.3 Literature Review

Recent research has increasingly focused on the use of technology to support adults who stutter, particularly through digital applications and therapy platforms that utilize Artificial Intelligence (AI) and Speech-to-Text (STT) technologies.

Three key studies have specifically addressed the main aspects upon which this project is based: the effectiveness of digital therapy, the use of STT for speech evaluation, and the lack of Arabic-language digital solutions for individuals who stutter.

First, the Systematic Review of Interventions for Adults Who Stutter (2020) highlighted that digital and remote (teletherapy) interventions produced results comparable to traditional in-person therapy among adults. The study found that technological tools help improve fluency, reduce stuttering frequency, and promote consistent practice by providing flexibility in time and location [1].

The findings in [1] support the concept of the current project, which aims to provide an accessible digital training tool that allows users to practice speech exercises regularly without attending therapy centers.

Second, in the Evaluating a Digital Speech Therapy App for Stuttering: A Pilot Validation Study (Eloquent), researchers evaluated a digital application that integrates Speech-to-Text (STT) technology to assess speech performance in real-time. The results showed significant improvement in fluency scores and a noticeable reduction in disfluencies after continuous use of the app [2].

The study in [2] reinforces the technical foundation of the proposed platform, which also relies on STT technology to automatically evaluate pronunciation accuracy and provide immediate feedback to users.

Finally, the study Using Technology in Intervention Services Provided to Arab Individuals Who Stutter (2023) examined how technology is being utilized in therapy for Arabic-speaking individuals who stutter. It identified a clear gap in the availability of Arabic-supported digital systems and emphasized the need for culturally and linguistically adapted tools for the Arab community [3].

In response, the proposed project focuses on developing an Arabic-only web-based platform designed to help adults who struggle with speech fluency interactively and track their progress in a user-friendly environment without requiring knowledge of a foreign language.

In summary, previous studies have demonstrated the effectiveness of digital therapy and the benefits of using Speech-to-Text technologies in speech training. However, they also reveal a major limitation—the absence of dedicated Arabic-language solutions. The proposed platform addresses this gap by combining automated STT-based evaluation with full Arabic language support in a free, accessible web platform tailored to Arabic-speaking adults who stutter.

2.4 Comparison Criteria Definition

To compare our proposed platform with other existing speech-therapy systems, several criteria were identified. These criteria focus on the main functional and linguistic differences between each system.

1. **Target Audience:** Determines whether the system is designed for adults or children.
2. **Arabic Language Support:** Checks if the system provides full Arabic interface and content.
3. **Use of Speech-to-Text (STT):** Evaluates whether the system integrates STT for automatic speech analysis and feedback.
4. **Free to Use:** Determines whether the system is freely accessible without subscriptions or paid sessions.

These criteria provide a simple and direct comparison framework that highlights how our proposed platform distinguishes itself from existing digital solutions.

2.5 Comparison results and the feasibility study

Existing System / Applications	Criterion				Remarks
	Target Audience	**Arabic Language**	**Uses STT**	**Free to Use**	
Stamurai [4]	Adults	No	No	No	
BeneTalk [5]	Adults	No	No	No	
TALAQA [6]	Children	Yes	Yes	Yes	
Our Proposed Platform	Adults	Yes	Yes	Yes	Arabic-only web-based platform for adults who stutter.

Table 2.2 **Comparison Results**

2.5.1 Comparison Results Summary:

The comparison shows that while TALAQA [6] integrates Arabic language and STT technology, it mainly targets children and uses a game-based structure. In contrast, our proposed platform focuses on adults who stutter, providing a fully Arabic, web-based environment that is simple, private, and accessible for continuous self-practice. It combines the advantages of STT technology, free access, and linguistic localization, which are not fully offered in other systems such as Stamurai [4] and BeneTalk [5].

2.5.1 Feasibility Study:

Technical Feasibility: The platform can be developed using standard web technologies such as HTML, CSS, JavaScript, and Firebase, integrated with Google Speech-to-Text API for real-time analysis. These technologies are reliable and well-documented, making implementation feasible.

Economic Feasibility: The project requires minimal financial resources. It relies on free or low-cost hosting and APIs, ensuring sustainability for both academic and long-term use.

Operational Feasibility: The platform is browser-based and requires no installation. Users can train from anywhere and at any time, providing high accessibility for adults with limited availability.

Time Feasibility: The project timeline is realistic and divided into multiple phases: planning, design, implementation, and testing. Each phase is achievable within the academic semester schedule, ensuring that development and evaluation are completed on time.

Legal and Ethical Feasibility: The project fully complies with ethical research standards. User data (such as speech recordings or progress logs) will not be shared or stored permanently without consent. The platform will not collect personal or sensitive information, and its use is voluntary and for educational purposes only, following standard privacy and ethical guidelines.

CHAPTER 3
ANALYSIS

3.1 Functional Requirements

No.	Functional Requirement	Description
1	User Registration	The system shall allow users to create a new account by entering basic information such as name, email, and password.
2	User Login and Logout	The system shall allow users to securely log in to their account and log out to end the session.
3	Display Speech Exercises	The system shall display Arabic words or sentences for users to practice based on their current level.
4	Speech-to-Text Evaluation (STT)	The system shall record the user's voice, convert it into text, and compare it with the target content to calculate pronunciation accuracy.
5	Instant Feedback	The system shall provide immediate feedback such as "Take a breath" or "Speak slowly" when pronunciation is incorrect.
6	Progress Tracking	The system shall track user performance, including correct answers and the current level.
7	Automatic Exercise Advancement	The system shall automatically move to the next exercise once the correct pronunciation threshold is reached.
8	Consultant Access to Shared Account	The system shall allow the speech consultant to view the patient's progress using the same account credentials.
9	Arabic Interface (RTL)	The system shall provide a fully Arabic user interface with right-to-left text alignment.

Table 3.1 Functional Requirements

3.2 Non-Functional Requirements

No.	Non-Functional Requirement	Description
1	Performance	The system shall process and display speech-to-text results quickly to ensure a smooth user experience.
2	Reliability	The system shall operate stably without unexpected crashes or interruptions during normal use.
3	Usability	The interface shall be clear, simple, and easy to use for adult users.
4	Compatibility	The system shall function properly on common devices and browsers without technical issues.
5	Security	The system shall protect user data and connections using appropriate security measures.
6	Maintainability	The platform must be designed for easy maintenance and updating, allowing quick integration of new features and efficient issue resolution.

Table 3.2 Non-Functional Requirements

3.3 UML DIAGRAMS

3.3.1 Global Use Case Diagram:

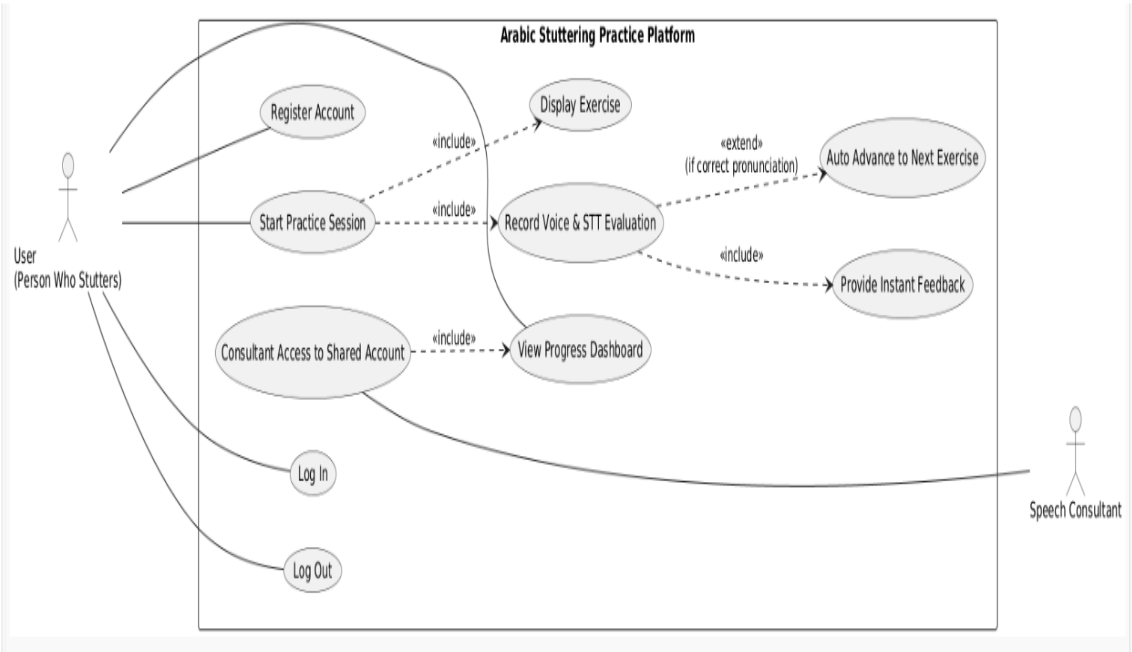


Figure 3.1 Use Case

3.3.2 Use Case Specification (Tabular Descriptions)

No.	UC1
Use Case Name	Register Account
Actors	User / Speech Consultant
Description	Allow both users and speech consultants to create a new account on the platform using an email address.
Pre-conditions	User or consultant not logged in; email not registered.
Post-condition	Account successfully created and stored.

***Table 3.3* Use Case Specification: Register Account**

No.	UC2
Use Case Name	Log In
Actors	User / Speech Consultant
Description	Allow users and consultants to log into their account.
Pre-conditions	A valid account must exist.
Post-condition	User or consultant successfully logs in.

Table 3.4 Use Case Specification: Log In

No.	UC3
Use Case Name	Log Out
Actors	User / Speech Consultant
Description	Allow users and consultants to log out securely.
Pre-conditions	User or consultant must be logged in.
Post-condition	Session is safely terminated.

Table 3.5 Use Case Specification: Log Out

No.	UC4
Use Case Name	Start Practice Session
Actors	User
Description	Begins a speech practice session for the current level.
Pre-conditions	User is logged in; exercises for the level exist.
Post-condition	Practice session initialized and ready for exercise flow.

Table 3.6 Use Case Specification: Start Practice Session

No.	UC5
Use Case Name	Display Exercise
Actors	User
Description	Displays Arabic words or sentences suitable for the user's current level.
Pre-conditions	Practice session started.
Post-condition	Exercise content displayed to the user.

Table 3.7 Use Case Specification: Display Exercise

No.	UC6
Use Case Name	Record Voice & STT Evaluation
Actors	User
Description	Records the user's voice and evaluates pronunciation using STT technology.
Pre-conditions	Microphone permission granted; exercise displayed.
Post-condition	Speech converted to text and evaluated with a score.

Table 3.8 Use Case Specification: Record Voice & STT Evaluation

No.	UC7
Use Case Name	Provide Instant Feedback
Actors	User
Description	Provides instant feedback only when the user's pronunciation is incorrect.
Pre-conditions	STT evaluation completed and result below threshold.
Post-condition	Feedback message displayed to guide user for retry.

Table 3.9 Use Case Specification: Provide Instant Feedback

No.	UC8
Use Case Name	Auto Advance to Next Exercise
Actors	User
Description	Automatically moves to the next exercise when the user's pronunciation is correct.
Pre-conditions	STT evaluation completed and pronunciation meets threshold.
Post-condition	System loads the next exercise or ends session if all completed.

Table 3.10 **Use Case Specification: Auto Advance to Next Exercise**

No.	UC9
Use Case Name	View Progress Dashboard
Actors	User / Speech Consultant
Description	Displays performance information such as number of attempts, completed levels, and progress trends.
Pre-conditions	Actor is logged in; progress data exists.
Post-condition	Dashboard displayed showing recent progress.

Table 3.11 Use Case Specification: View Progress Dashboard

No.	UC10
Use Case Name	Consultant Access to Shared Account
Actors	User / Speech Consultant
Description	Both the user and consultant share the same account; the consultant can log in with the same credentials to review or perform exercises as the patient.
Pre-conditions	Consultant authenticated using shared account credentials.
Post-condition	Consultant can access and interact with patient’s exercises and progress.

Table 3.12 Use Case Specification: Consultant Access to Shared Account

3.3.3 Class Diagram

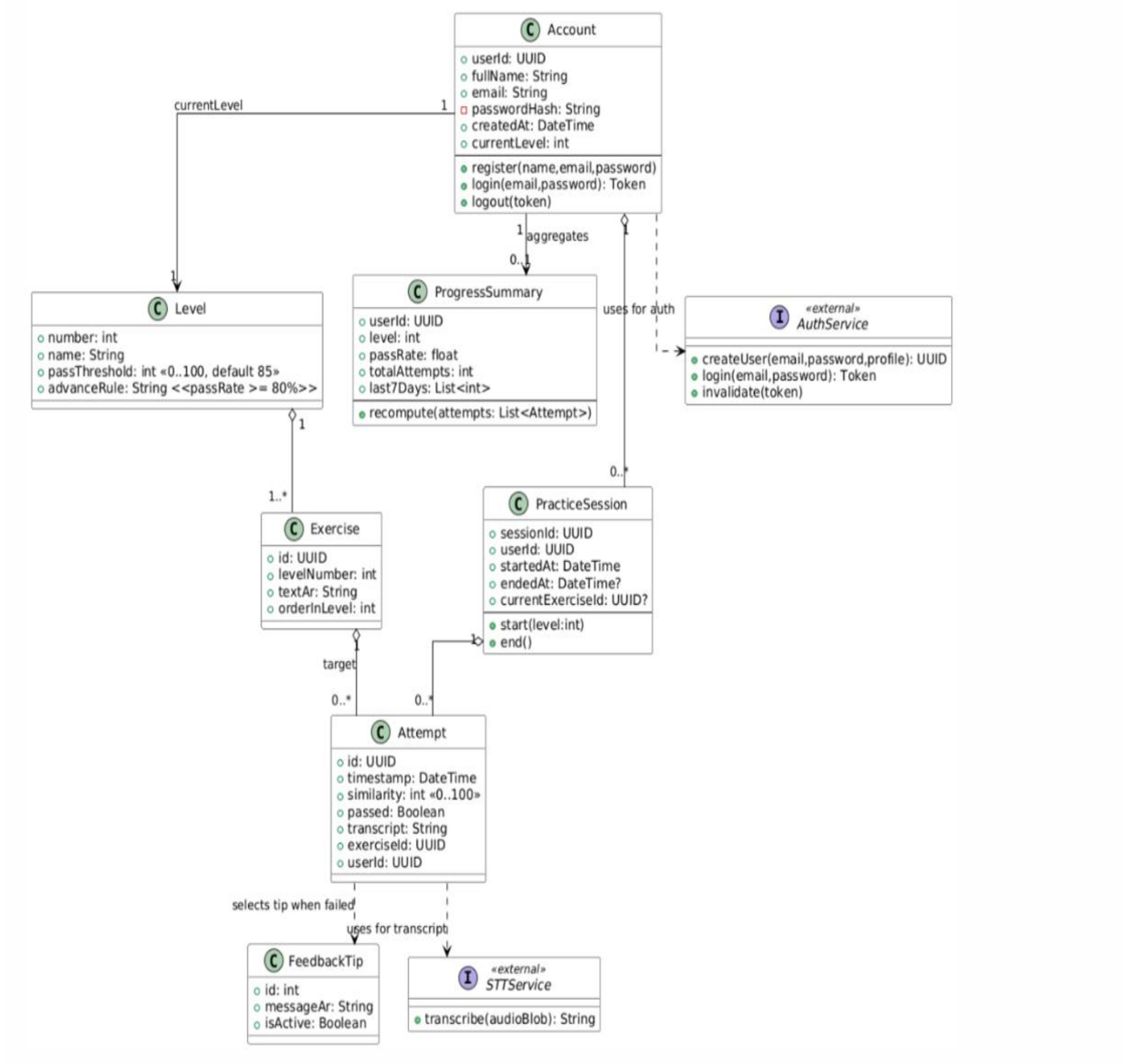


Figure 3.2 Class Diagram

3.3.4 Sequence diagrams:

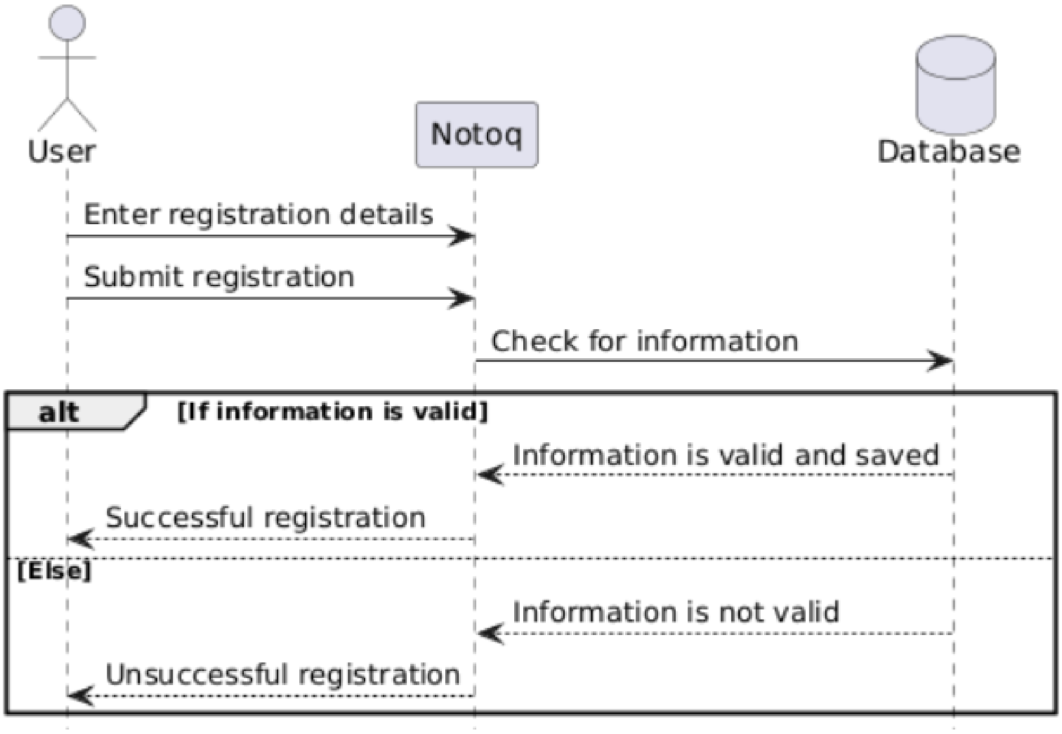


Figure 3.3 Register Account

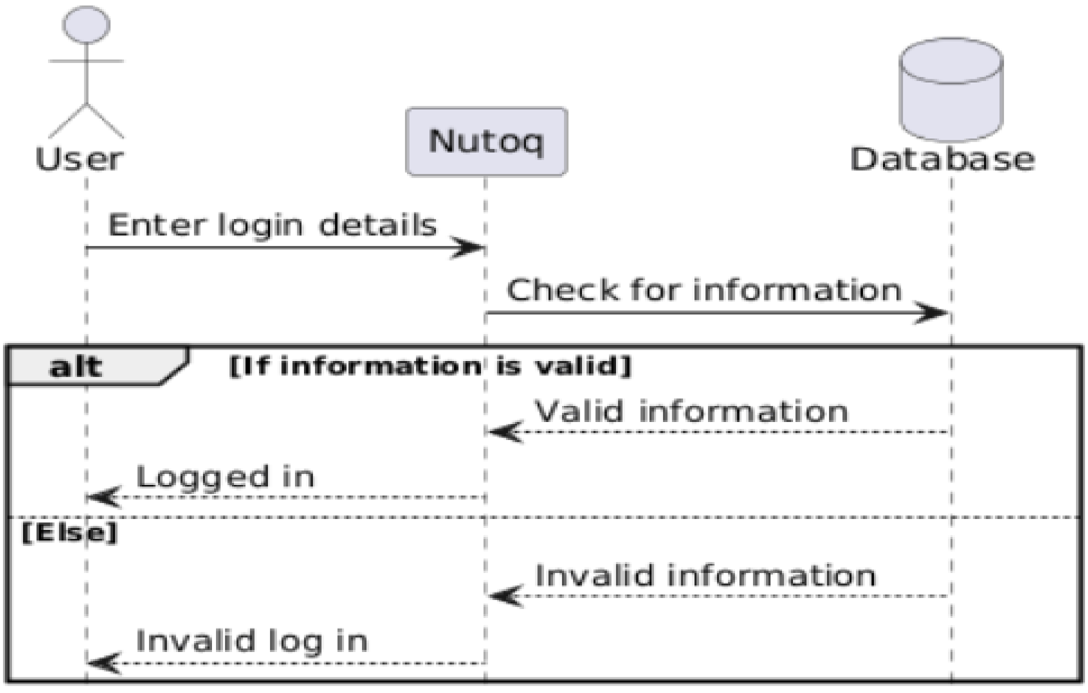


Figure 3.4 Log In

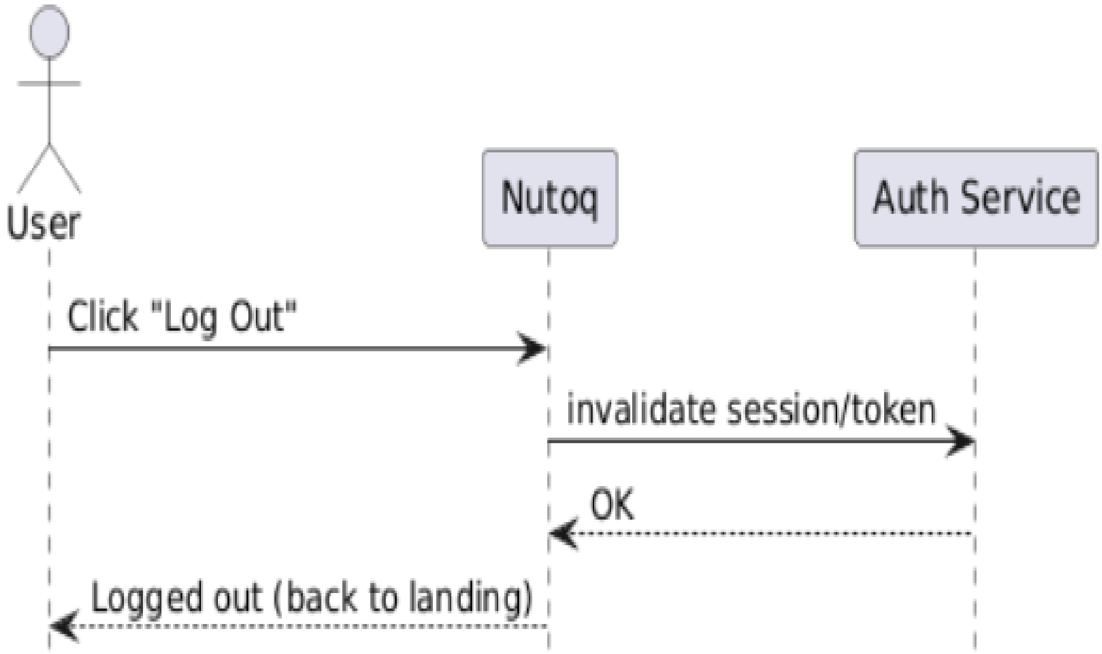


Figure 3.5 **Log Out**

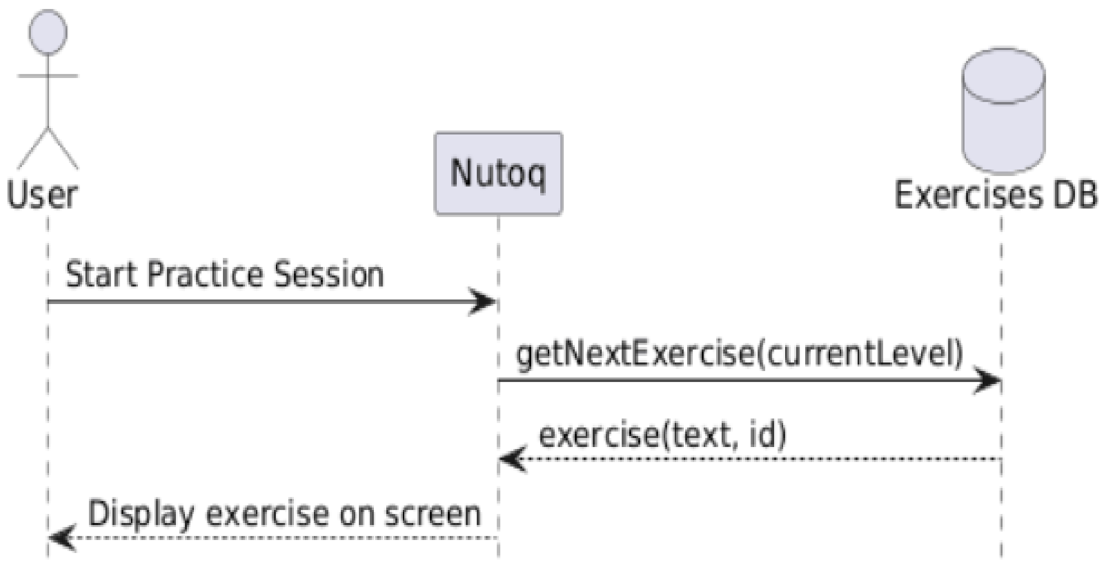


Figure 3.6 **Start Practice Session**

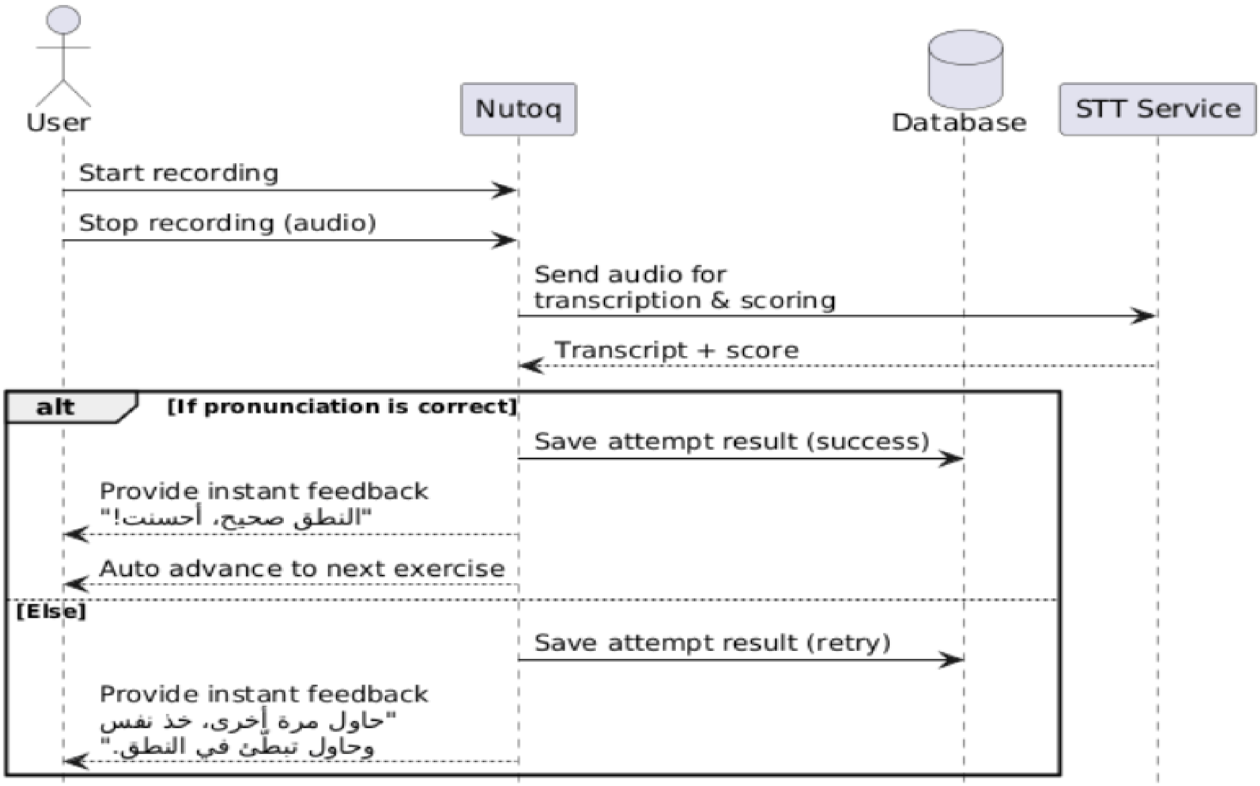


Figure 3.7 Record Voice & STT Evaluation

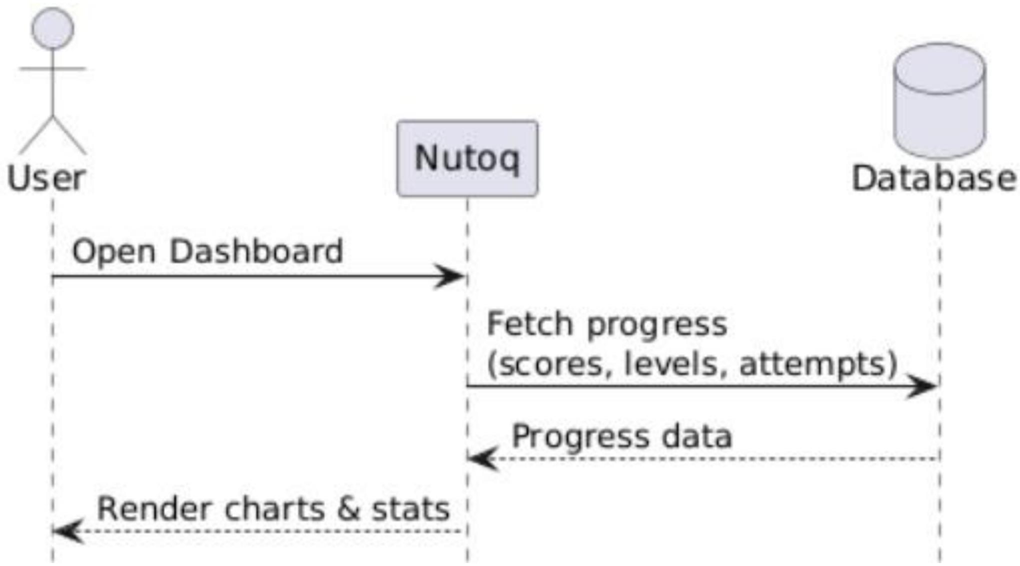


Figure 3.8 View Progress Dashboard

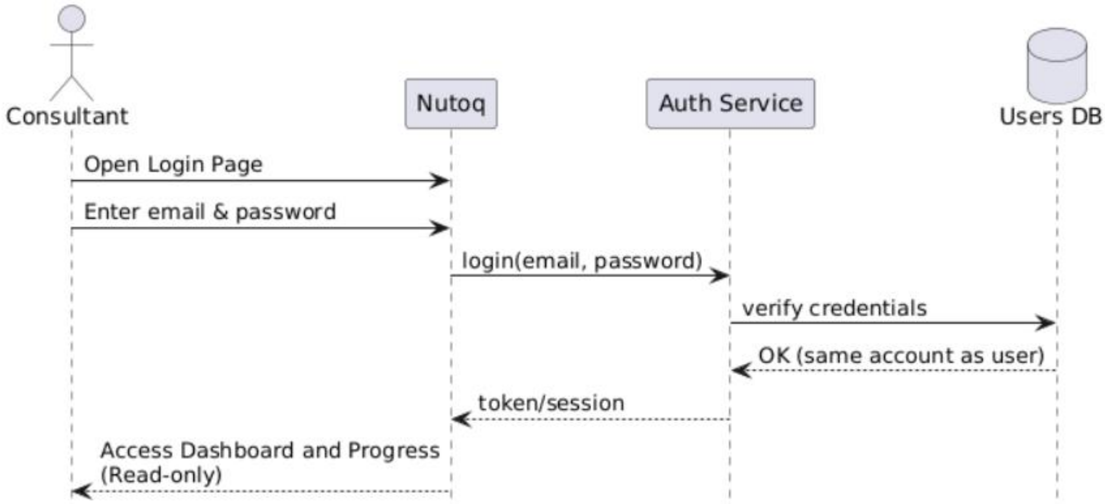


Figure 3.9 Consultant Access to Shared Account

3.3.5 Context Diagram

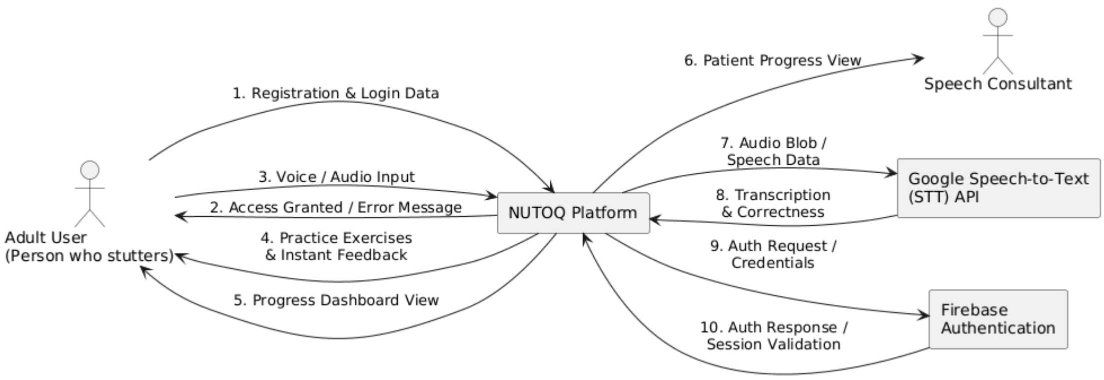


Figure 3.10 Context Diagram

3.4 Data collection instruments

We developed a Google form and distributed it to speech therapy specialists to gather essential information that will aid us in tailoring our platform according to their instructions. In the form, we asked them about how effective our platform could be. These questions were designed to delve deeper into their experiences and needs, which will inform the development of more targeted features for proposed platform.

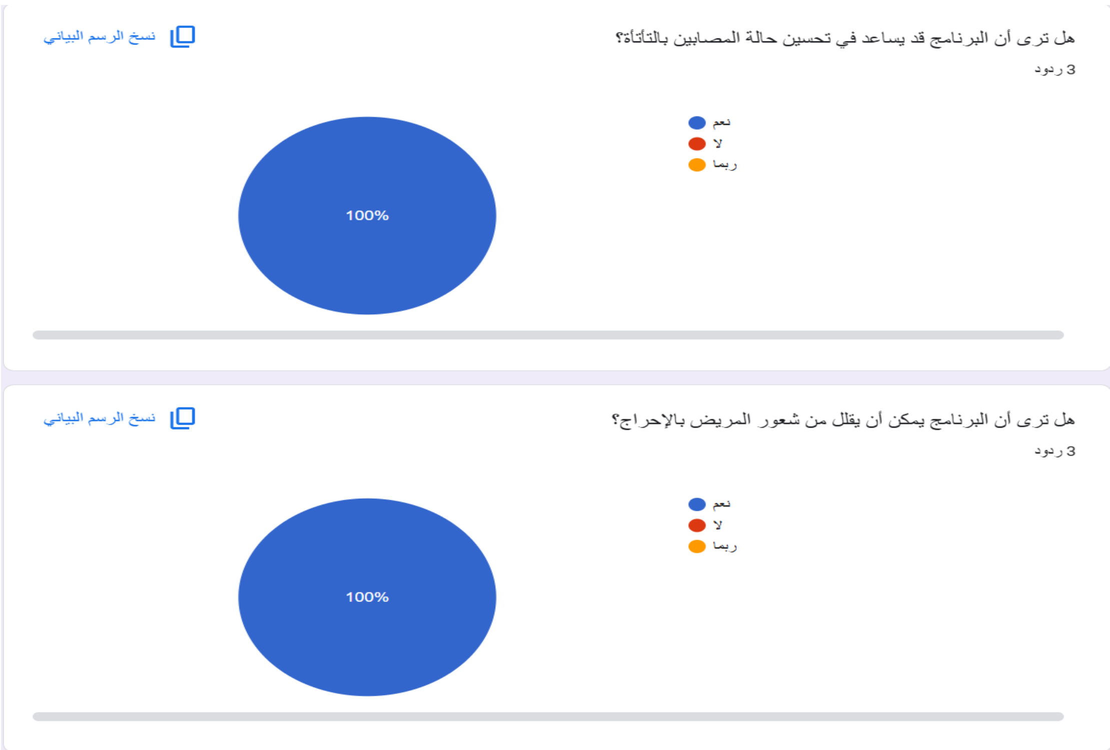


Figure 3.11 First and Second Question in Survey

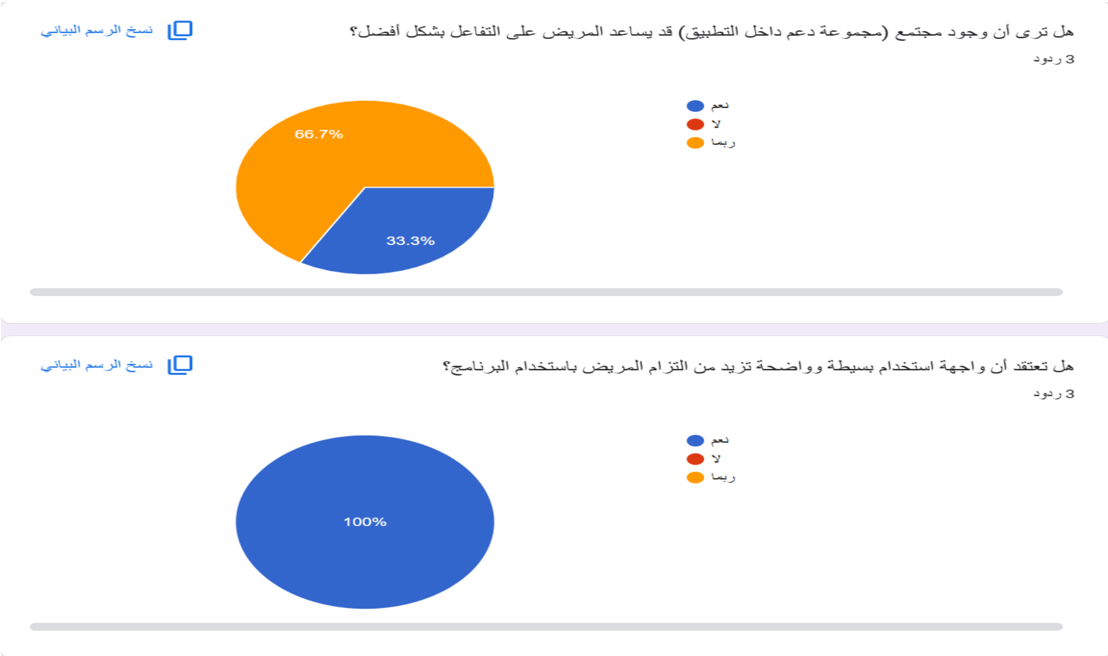


Figure 3.12 **third and Fourth Question in Survey**

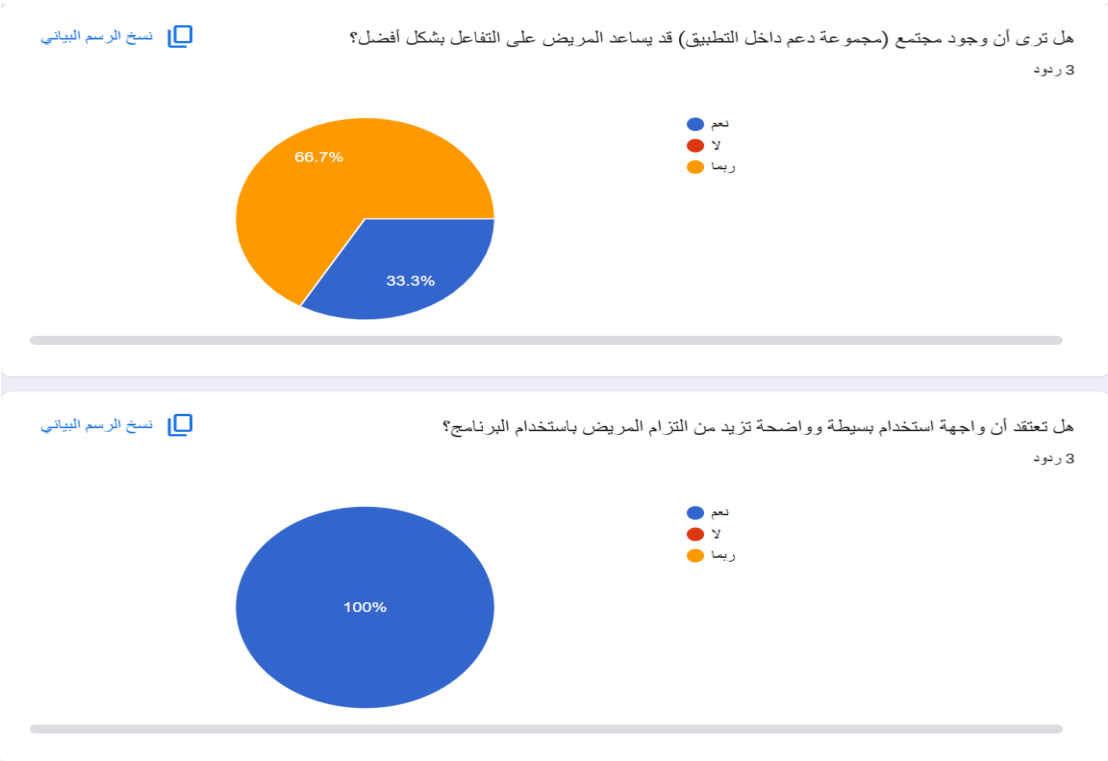


Figure 3.13 Fifth and Sixth Question in Survey

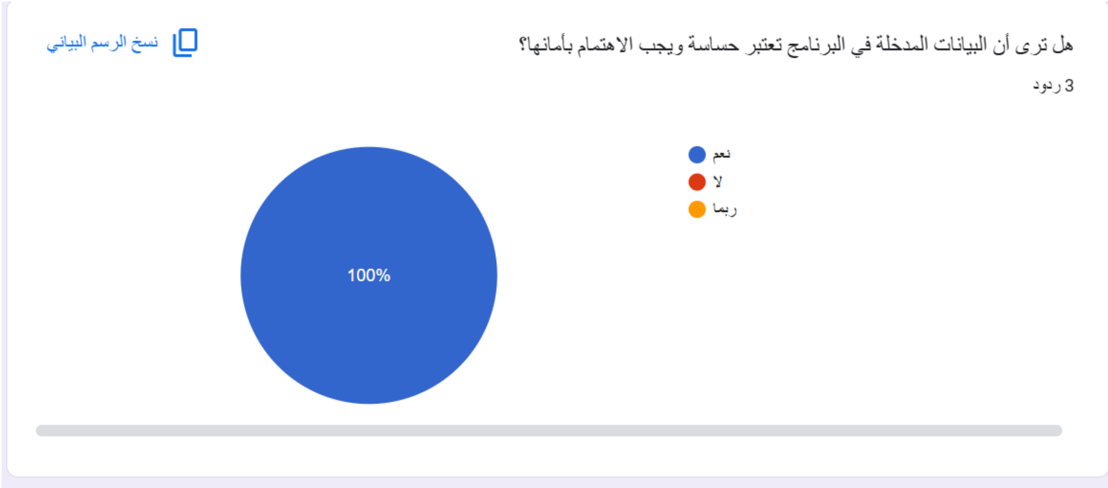


Figure 3.14 Seventh Question in Survey

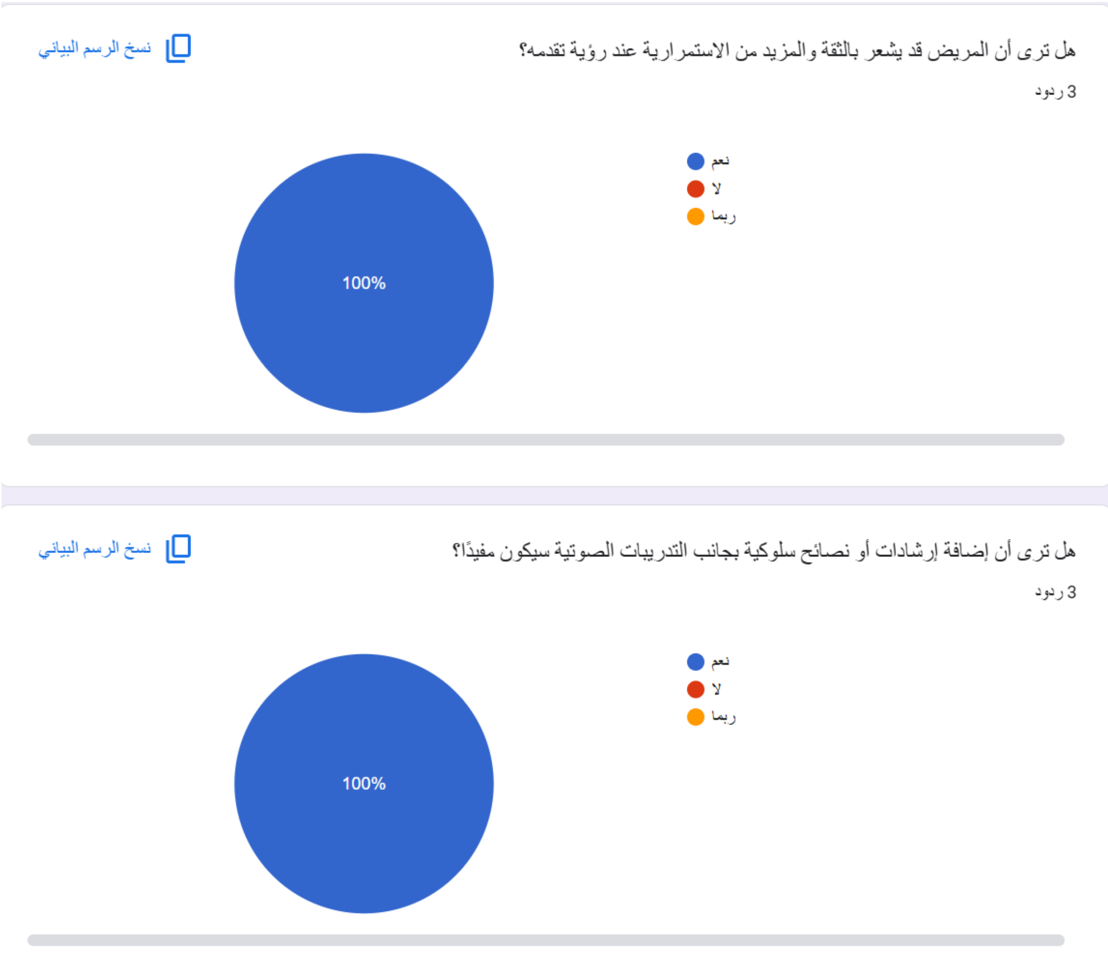


Figure 3.15 Eight and ninth Question in Survey

CHAPTER 4
DESIGN

4.1 Architecture Overview

In general, we design the Arabic stuttering practice platform using a Layered Architecture, organizing it as a web application that users access through a standard internet browser. This architectural choice targets Arabic-speaking adults who stutter and need a simple way to practice speech at home using interactive, text-based exercises with automatic feedback.

The layered architecture keeps the platform modular, with a clear separation of concerns between the presentation layer, the application layer, and the data layer. Users interact with the platform through an Arabic web interface, where they can start practice sessions, record their voice, and view their progress. The application layer manages the practice flow and communicates with an external Speech-to-Text (STT) service to evaluate pronunciation. The data layer stores users' accounts, exercises, and progress so that both the user and the consultant can review performance over time.

This structure improves maintainability and scalability, because changes in one layer (for example, replacing the STT provider or modifying the database structure) can be made with minimal impact on the other layers.

4.2 Architectural Pattern

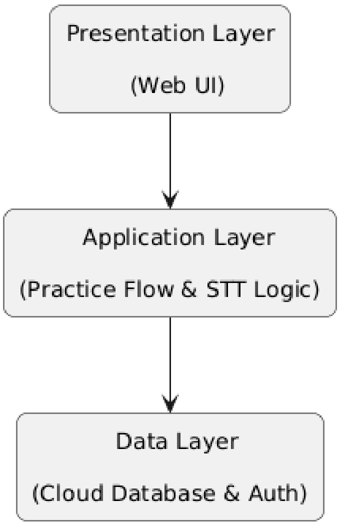


Figure 4.1 **Layered Architectural Pattern**

4.3 User Interface Design

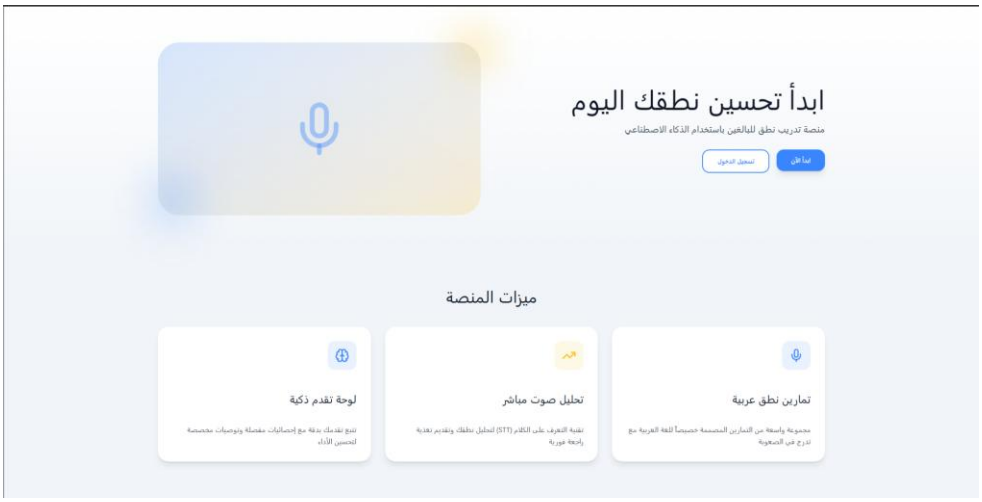


Figure 4.2 **Public Home Page**

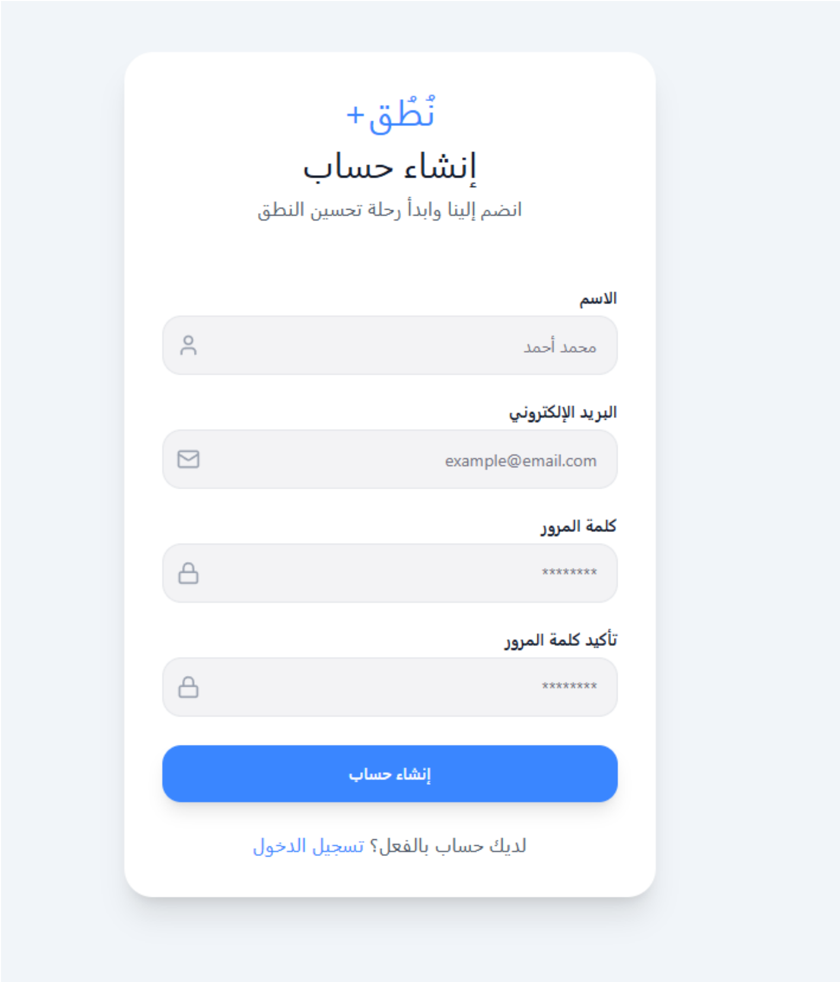


Figure 4.3 Registration Page

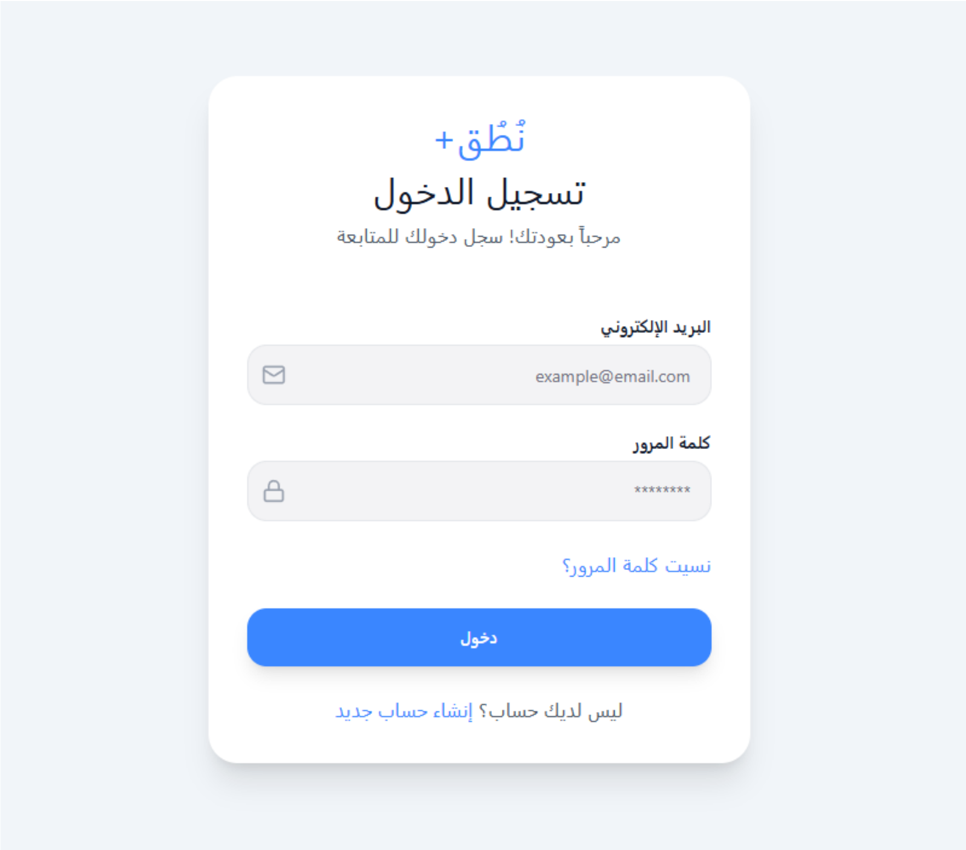


Figure 4.4 Login Page

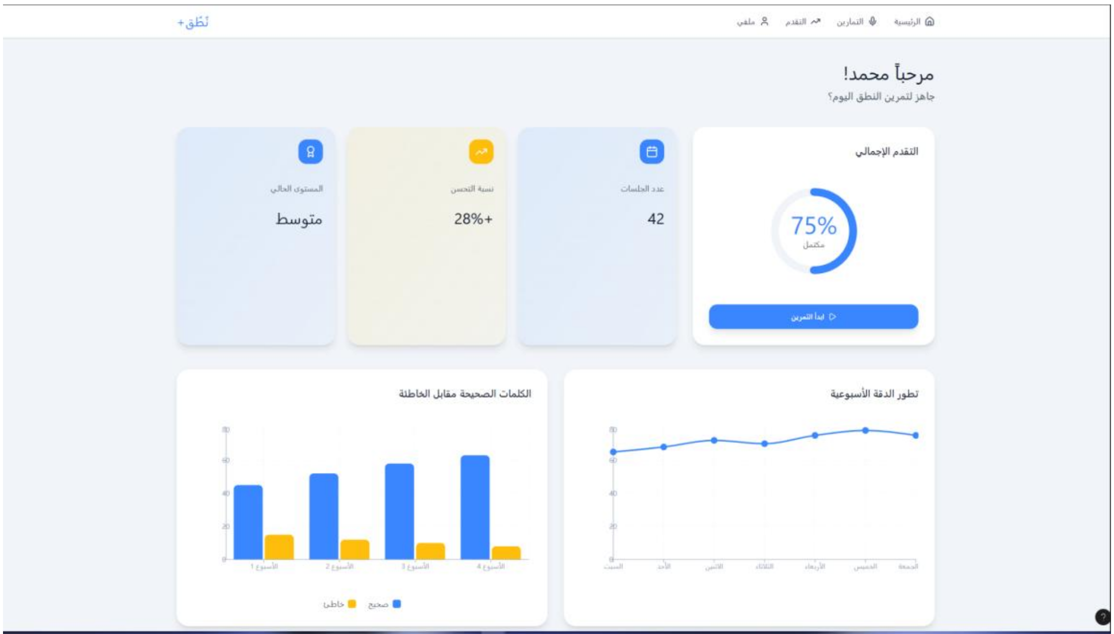


Figure 4.5 Progress Dashboard

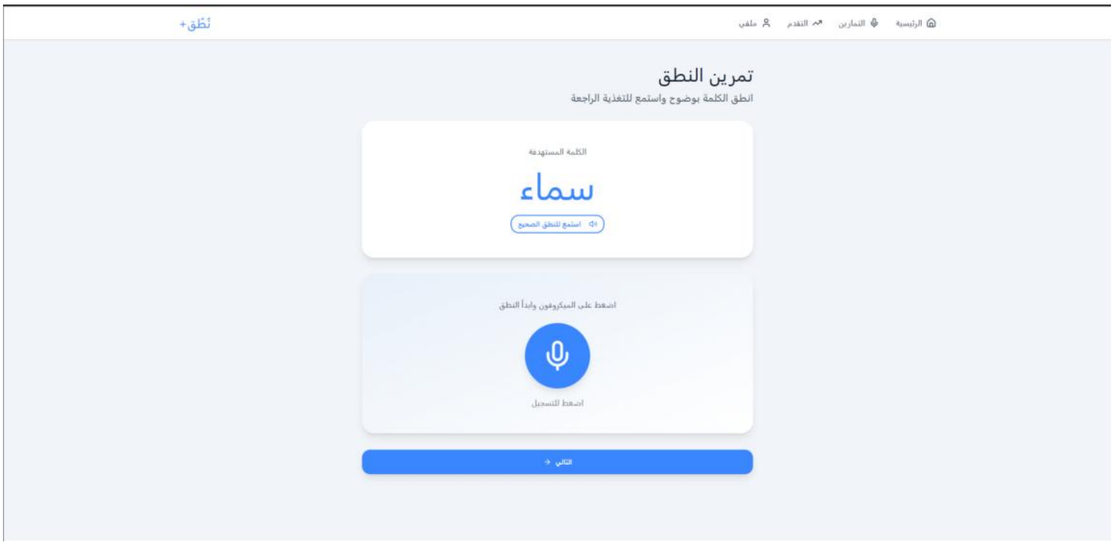


Figure 4.6 Pronunciation Practice Srceen

References

[1] Systematic Review of Interventions for Adults Who Stutter, 2020.

[2] Evaluating a Digital Speech Therapy App for Stuttering (Eloquent), 2025.

[3] Using Technology in Intervention Services Provided to Arab Individuals Who Stutter, 2023.

[4] https://www.stamurai.com/

[5] https://www.superpenguin.com/

[6] TALAQA, 2025.